

Evapo(transpi)ration

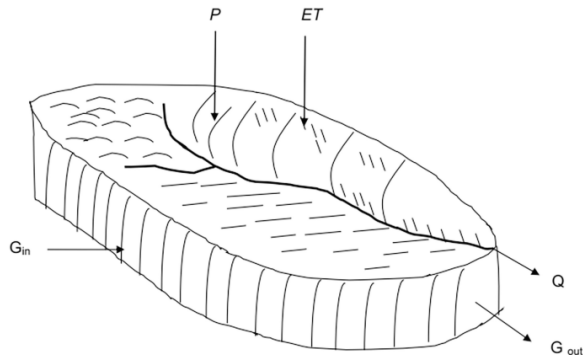
Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- In PS #3, "log frequency" should be "log exceedance probability" in problem 1.5.
- PS #3 deadline pushed to Tuesday
- No class next Tuesday!

Regional Water Balance

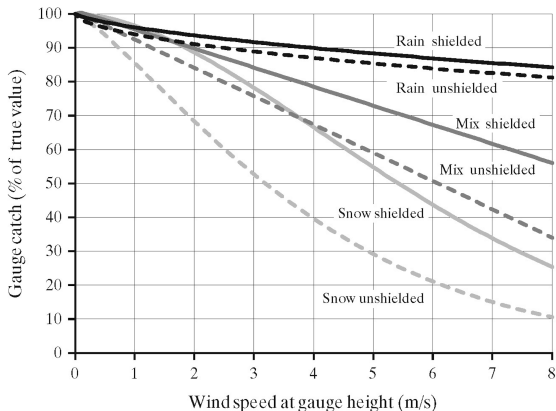
$$\hat{P} + G_{in} - (Q + ET + G_{out}) = \Delta S$$



Where:

- P = Precipitation
- ET = Evapotranspiration
- Q = Stream outflow
- G_{in} = Ground water in
- G_{out} = Ground water out

Precip Uncertainty



Sources:

- gage catch (10-90%)
- interpolation
- length of timeseries (stationarity assumption)
- unmeasured precip

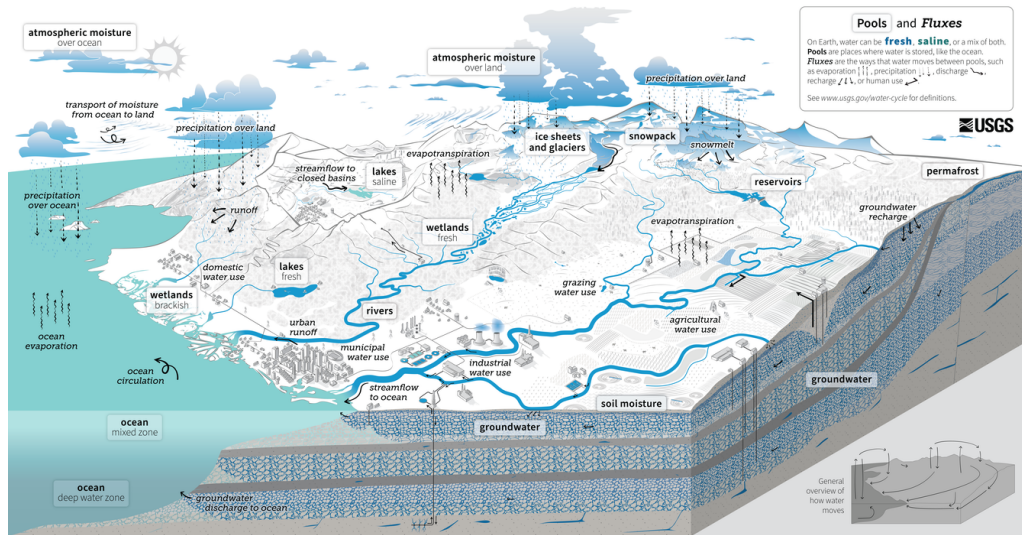
Approaches:

- integrate/validate w/ other data sources
- ensemble/Monte carlo
- modeling

Learning Objectives

- 1 Evaporation processes
- 2 Methods for estimating potential evaporation
- 3 Transpiration process and estimation methods Interception (loss)

Water Cycle



Evapotranspiration

Evapotranspiration (ET) is collective term for all solid/liquid phase water at land surface that becomes atmospheric water vapor.

It comprises **evaporation**: direct vaporization of water from open water surfaces and bare soil (can also have “sublimation” from snow/ice)

and **transpiration**: vapor flux via leaves of vegetation.

Importance of ET

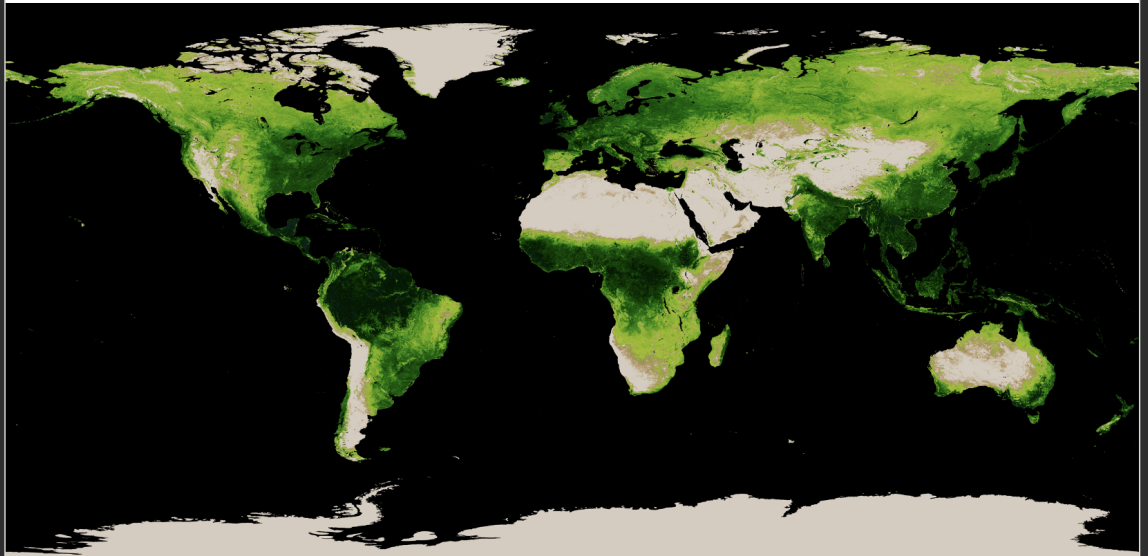
- 1 Globally, more than half (62%) of precipitation (P) becomes ET, the majority of which is evapotranspiration (97%) from land surfaces, with 3% occurring as open-water evaporation.
- 2 The difference between P and ET is water availability for human use.
- 3 Irrigated agricultural is the largest consumptive use of water, and understanding of transpiration is crucial for efficient application.
- 4 ET controls the moisture content of soil, which in turn impacts infiltration and groundwater recharge.

Actual vs Potential ET

Potential ET (PET) is the total climatic evaporative demand on the land surface, and is typically much larger than **actual ET** (AET), the actual rate of solid/liquid phase water that becomes water vapor.

AET is $<$ than PET due to either a limitation in water-availability (i.e., "water-limited") or energy availability (i.e., "energy-limited"). The former occurs in arid regions where $PET > P$, and the latter occurs in humid regions where $P > PET$.

Normalized Difference Vegetation Index (NDVI)



Challenge

- Direct measurement of actual evapotranspiration is much more difficult (and expensive) than precipitation or streamflow.
- Closest readily available (“easy”) measurement network are evaporation pans, which only measure open water (potential) evaporation.
- Therefore need to develop models for estimating ET from more readily available observations.

Physics of Evaporation

Evaporation is a diffusive process that follows Fick's first law: particles move from high concentration to low concentration proportional to the concentration gradient.

$$E = K_E \cdot v_a \cdot (e_s - e_a)$$

- E is evap rate (LT^{-1})
- e_s and e_a are vapor pressure of evap surface and air ($\text{ML}^{-1}\text{T}^{-2}$). Vapor pressure at the evaporation surface is the saturation vapor pressure (e^*)
- v_a is wind speed (LT^{-1})
- K_E is coefficient representing efficiency of vertical transport of water by turbulent eddies

Physics of Evaporation

Evaporation is a diffusive process that follows Fick's first law: particles move from high concentration to low concentration proportional to the concentration gradient.

$$E = K_E \cdot v_a \cdot (e_s - e_a)$$

"Evaporation is proportional to the product of the near-surface vapor pressure difference and wind speed"

Temperature

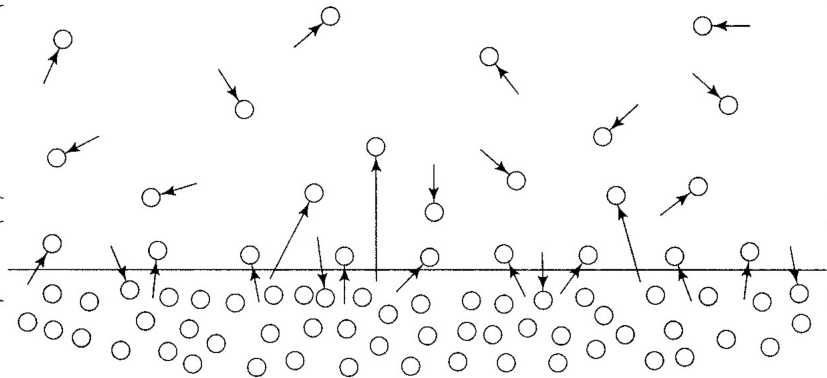
T_a

T_s

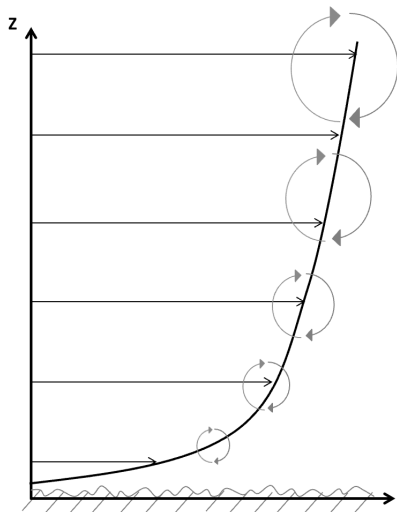
Vapor pressure

$e_a \leq e^*(T_a)$

$e^*(T_s)$



Wind Speed



The vertical profile of horizontally flowing wind is affected by friction with the land surface, producing turbulent eddies that transport latent and sensible heat towards or away from the surface and a logarithmic increase in speed with height above the surface.

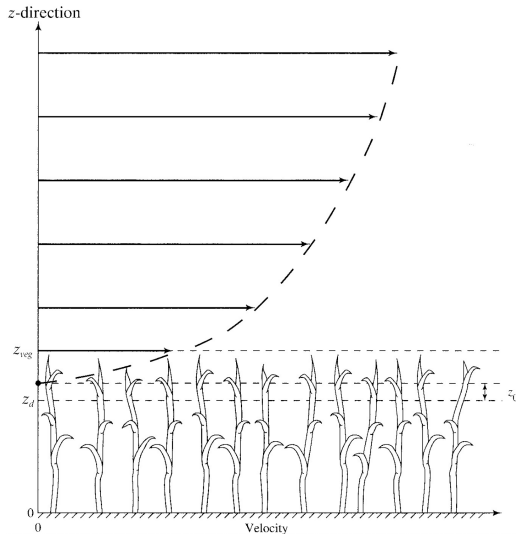
Physics of Evaporation

K_E (having units of L^2F^{-1}) can be estimated as:

$$K_E = \frac{0.622 \cdot \rho_a}{p \cdot \rho_w} \cdot \frac{1}{6.25 \cdot \left[\ln\left(\frac{z_m - z_d}{z_0}\right) \right]} \quad (1)$$

where p is atmospheric pressure, z_m is the measurement height (typically 2 m), z_d is the zero-plane displacement height (height above ground where mean wind speed would theoretically be 0 due to large obstacles), and z_0 is the roughness height (or height above surface where wind speed would be zero due to small-scale roughness).

Zzz's



wind speed (v_z) = 0 at $z = z_d + z_0$, which is called the virtual surface.

$$v_z = 0 \text{ at } z = z_d + z_0$$

$$z_d = .7 \cdot z_{veg}$$

$$z_0 = .1 \cdot z_{veg}$$

therefore

$$z = 0.8 \cdot z_{veg}$$

Latent and Sensible Heat Exchange

Recall latent heat is released due to a phase change (evaporation/condensation) calculated as:

$$LE = \rho_w \cdot \lambda_v \cdot E$$

where ρ_w is density of water, E is evaporation rate and λ_v is latent heat of vaporization, calculated as:

$$\lambda_v = 2.50 - 2.36 \times 10^{-3} \cdot T$$

Sensible Heat Exchange

Sensible heat exchange results in a temperature change and occurs via turbulent diffusion in the near surface.

$$H = K_H \cdot v(z_m) \cdot [T_s - T(z_m)]$$

where K_H is the sensible heat transfer coefficient. Look familiar?

Bowen Ratio

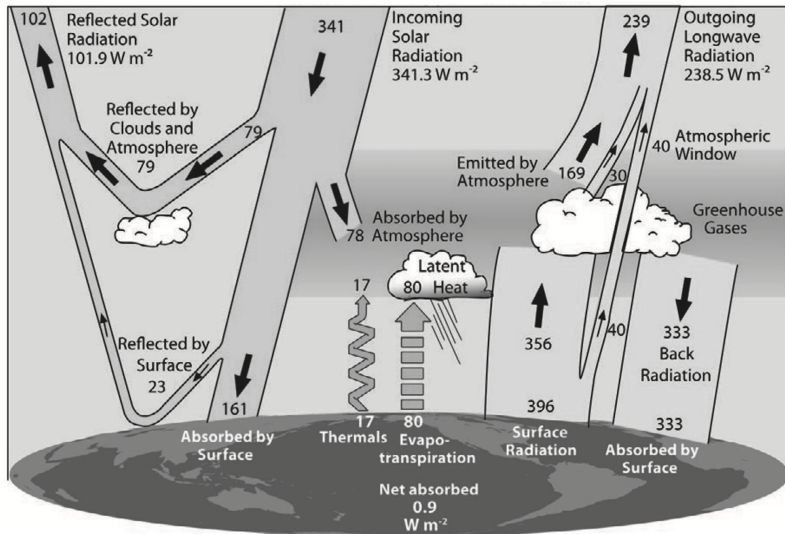
Ratio of sensible to latent heat exchange:

$$B = \frac{H}{LE} \quad (2)$$

H and LE are considered positive when directed towards the atmosphere and negative when directed towards the surface.

Type of surface	Range of Bowen ratios
Deserts	>10.0
Semi-arid landscapes	2.0-6.0
Temperate forests and grasslands	0.4-0.8
Tropical rainforests	0.1-0.3
Tropical oceans	<0.1

Energy Balance



Energy Balance

For an evapotranspiring body during time period Δt :

$$LE = K + L - G - H + A_w - \frac{\Delta Q}{\Delta t}$$

- LE: latent heat ^{exchange} ~~of vaporization~~
- K: net shortwave radiation
- L: net longwave radiation
- G: net output via conduction to ground
- H: net output from sensible heat exchange with atmosphere
- A_w : water-advected energy (only applicable to lakes)
- $\frac{\Delta Q}{\Delta t}$: change in amount of heat stored in water body per unit area per unit time (only applicable to lakes)

ET Classification

- 1 Type of surface (open water, bare soil, leaf, canopy, crop, land region)
- 2 Water Availability (unlimited or limited)
- 3 Stored energy use
- 4 Water-advected energy use

The last two are only applicable to lake evaporation.

Potential Evaporation

Potential or free-water evaporation is the rate of evaporation that would occur from an extended, open-water surface under current meteorological (vapor pressure/wind) conditions without heat-storage or water-advected energy effects.

Lake evaporation is all of the above and includes water-advected and heat storage effects. We won't be talking too much more about lakes... sorry.

Quantification of Potential Evaporation

- 1 Pan Evaporation
- 2 Water Balance
- 3 Mass Transfer
- 4 Energy Balance
- 5 Penman/Combination

Why so many approaches??

Pan Evaporation

Only direct measurement of evaporation.

$$E = P - [S_2 - S_1]$$

where E is evaporation, P is precipitation, and S is storage at beginning (S_1) and end (S_2) of time period.



Pan Evaporation

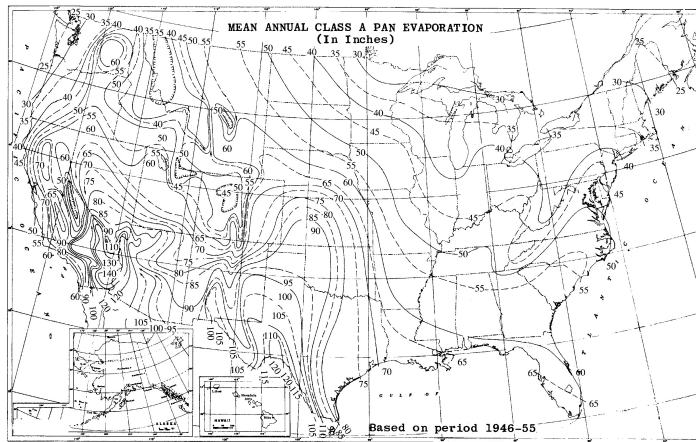


Figure 6.6 Average annual pan (= free-water-surface) evaporation (in/yr $\cdot$ 25.4 = mm/yr) for the 48 contiguous United States based on data for 1946–1955 (map provided by US National Weather Service).

WRRC
NOAA

Pan Coefficients

Radiation on the pan sidewalls and the lower heat-storage capacity compared to a body of water means that pan evap measurements usually need to be corrected (downwards) using a **pan coefficient** (ratio of "true" to pan evaporation). There are also empirical approaches based on air temperature, see textbook.

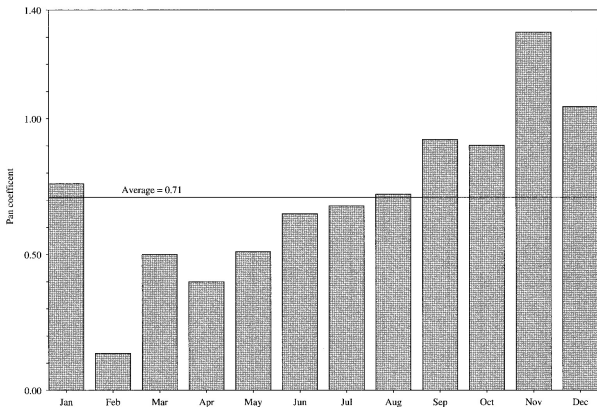


Figure 6.5 Monthly Class-A pan coefficients at Lake Hefner, Oklahoma, June 1950–May 1951 [data from Harbeck et al. (1954)].

Water-Balance Approach

Re-arrange water balance to solve for E:

$$Q_{in} + P + G_{in} - (Q_{out} + E + G_{out}) = \Delta S$$
$$E = Q_{in} + P + G_{in} - Q_{out} - G_{out} - \Delta S$$

Limitations:

We don't often have good data in these components

Mass-Transfer Approach

$$E = K_E \cdot v_a \cdot (e_s - e_a)$$

Recall:

$$e_s = e_s^* = 0.611 \cdot \exp \frac{17.3 \cdot T_s}{T_s + 273.3}$$

Solve for K_E using published (or estimated) values of z_m , z_0 , and z_d , and measured values of humidity, wind speed, and air and water surface temperature.

Limitations: no data on T_s

Energy-Balance Approach

Re-arrange energy balance to solve for E based on relationship to latent heat exchange (LE) and include only terms for surface heat exchange:

$$\frac{LE}{\rho_w \cdot \lambda_v} = \frac{\rho_w \cdot \lambda_v \cdot E}{\rho_w \cdot \lambda_v} \leftarrow LE = K + L - G - H + \cancel{A_w} - \cancel{\frac{\Delta Q}{\Delta t}}$$
$$E = \frac{K + L - G - H}{\rho_w \cdot \lambda_v}$$

Limitations:

Both $H + L$ depend on T_s

Penman/Combined Approach

Penman showed (in 1948) that the mass-transfer and energy-balance approaches could be combined to produce an equation that did not require T_s (something that is difficult to measure for natural, free water surfaces).

$$E = \frac{\Delta \cdot (K + L) + \gamma \cdot K_E \cdot \rho_w \cdot \lambda_v \cdot u(z_m) \cdot e^*(z_m) \cdot [1 - RH(z_m)]}{\rho_w \cdot \lambda_v \cdot (\Delta \cdot \gamma)} \quad (3)$$

Looks quite involved, but makes use of generally available data (there are numerous modifications available to address data limitations, see Dingman textbook) and performs quite well compared to pan evaporation estimates. It is therefore the "standard" approach for modeling potential evaporation.

Combined and Pan Evap Comparison

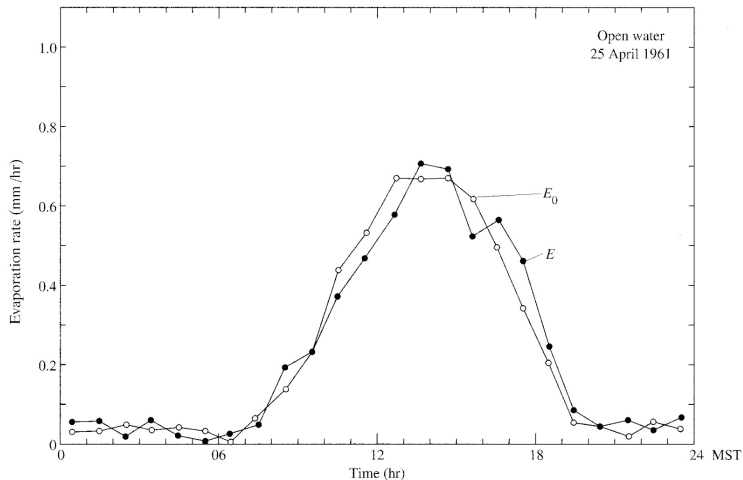


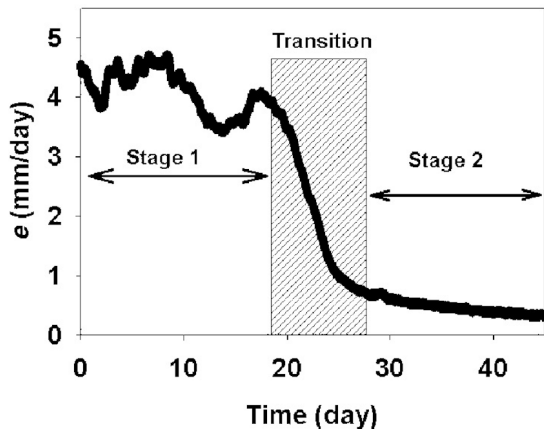
Figure 6.3 Comparison of hourly evaporation rates determined from measurements in a shallow pan (E) with those computed via the combination approach [E_0 , equation (6.36)] at Tempe, Arizona [Van Bavel (1966)]. Potential

Bare-Soil Evaporation

>1/3rd of earth's land surface supports little or no vegetation, making bare soil evaporation globally significant

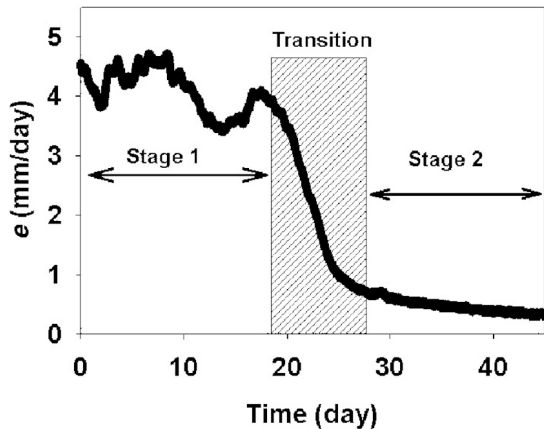
Following infiltration, evaporation from bare soil typically follows 3 stages:

- 1 atmosphere-controlled stage
- 2 transition
- 3 soil-controlled stage



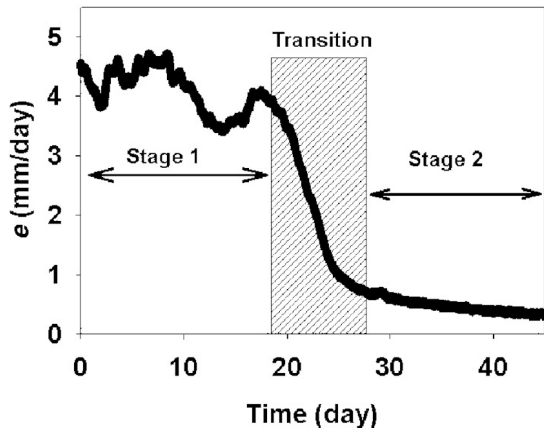
Atmosphere-controlled Stage

E_1 is determined by the surface energy balance and occurs at or near the rate of potential evaporation (see previous slides for quantification approaches).



Transition Stage

As the soil dries, a drying front forms and becomes hydraulically disconnected from surface. E declines asymptotically, approaching a constant-rate.

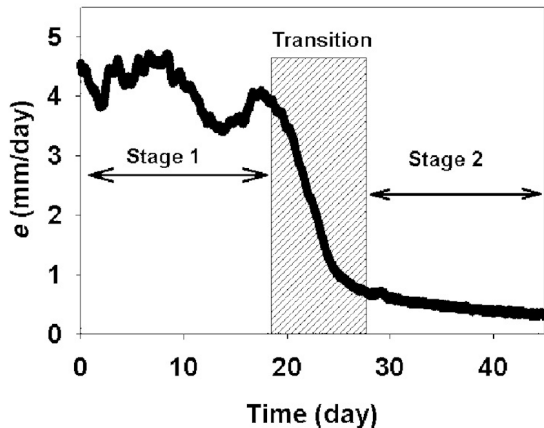


Soil-controlled Stage

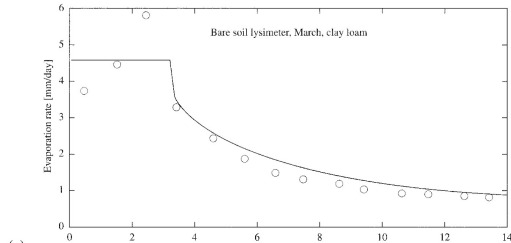
E_2 is determined by rate at which water is conducted in response to potential gradients induced by upward-decreasing soil water contents.

Estimated based on average stage 1 evaporation rate (E_1), stage 1 time (t_1) and total time elapsed (t):

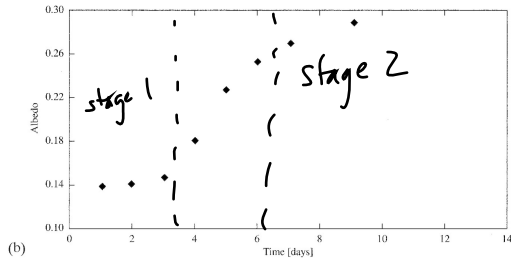
$$E_2 = \bar{E}_1 * \left(\frac{8}{\pi^{1/2}} \right) \cdot \frac{t_1}{t} \quad (4)$$



Bare Soil Evap



(a)



(b)

t_c determined
from albedo!